

5. Addition of Two 16-bit Data

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code in the center pane is as follows:

```
1 LDA 8050H
2 MOV B,A
3
4 LDA 8052H
5 ADD B
6 STA 8054H
7
8 LDA 8051H
9 MOV B,A
10
11 LDA 8053H
12 ADC B
13 STA 8055H
14
15 MVI A,00H
16 ACI 00H
17 STA 8056H
18
19 HLT
```

The registers window on the left shows the following values:

Register	Value
A	00
BC	12 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 1E
SP	FF FF
Int-Reg	00

The flag window shows the following values:

Flag	Value
S	0
Z	1
AC	0
P	1
C	0

The memory window on the right shows the following data:

Address (Hex)	Address	Data
8050	32848	52
8051	32849	18
8052	32850	120
8053	32851	86
8054	32852	172
8055	32853	104
8056	32854	0
8057	32855	0
8058	32856	0
8059	32857	0
805A	32858	0
805B	32859	0
805C	32860	0
805D	32861	0

The assembler message window at the bottom shows the message: "Program assembled successfully".

6. Subtraction of Two 16-bit Data

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code in the center pane is as follows:

```
1 LHLD 8050H
2 XCHG
3 LHLD 8052H
4 MOV A,L
5 SUB E
6 MOV L,A
7 MOV A,H
8 SBB D
9 MOV H,A
10 SHLD 8054H
11 HLT
```

The registers window on the left shows the following values:

Register	Value
A	BB
BC	00 00
DE	56 78
HL	BB BC
PSW	00 00
PC	42 11
SP	FF FF
Int-Reg	00

The flag window shows the following values:

Flag	Value
S	1
Z	0
AC	0
P	1
C	1

The memory window on the right shows the following data:

Address (Hex)	Address	Data
8050	32848	120
8051	32849	86
8052	32850	52
8053	32851	18
8054	32852	188
8055	32853	187
8056	32854	0
8057	32855	0
8058	32856	0
8059	32857	0
805A	32858	0
805B	32859	0
805C	32860	0
805D	32861	0

The assembler message window at the bottom shows the message: "Program assembled successfully".

7. Multiplication of Two 16-bit Numbers (Repeated Addition Method)

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window displays the assembly code for a program that multiplies two 16-bit numbers (8050H and 0000H) using the repeated addition method. The program is assembled successfully, and the memory window shows the data stored at various addresses.

Registers:

Register	Value
A	00
BC	00 00
DE	00 0A
HL	00 28
PSW	00 00
PC	42 25
SP	00 00
Int-Reg	00

Flags:

Flag	Value
S	0
Z	1
AC	0
P	1
C	0

Assembly Code:

```
1 LXI H,8050H
2 MOV E,M
3 INX H
4 MOV D,M
5 INX H
6 MOV C,M
7 INX H
8 MOV B,M
9 MOV B,M
10 LXI H,0000H
11 LXI SP,0000H
12 LOOP: MOV A,B
13 ORA C
14 JZ STORE
15 DAD D
16 DCX B
17 JMP LOOP
18 STORE: SHLD 8060H
19 XRA A
20 STA 8062H
21 STA 8063H
22 HLT
```

Memory Window:

Address (Hex)	Address	Data
8060	32864	40
8061	32865	0
8062	32866	0
8063	32867	0
8064	32868	0
8065	32869	0
8066	32870	0
8067	32871	0
8068	32872	0
8069	32873	0
806A	32874	0
806B	32875	0
806C	32876	0
806D	32877	0

Assembler Message:

```
0 Program assembled successfully
```

8. Division of Two 16-bit Numbers

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window displays the assembly code for a program that divides two 16-bit numbers (8050H and 0000H) using the repeated subtraction method. The program is assembled successfully, and the memory window shows the data stored at various addresses.

Registers:

Register	Value
A	00
BC	00 05
DE	00 00
HL	00 04
PSW	00 00
PC	42 30
SP	FF FF
Int-Reg	00

Flags:

Flag	Value
S	1
Z	0
AC	0
P	0
C	1

Assembly Code:

```
1 LXI H,8050H
2 MOV E,M
3 INX H
4 MOV D,M
5 INX H
6 MOV C,M
7 INX H
8 MOV B,M
9 MOV B,M
10 LXI H,0000H
11 LOOP: MOV A,D
12 CMP B
13 JC DONE
14 SUBTRACT: MOV A,E
15 SUB C
16 MOV E,A
17 MOV A,D
18 SBB B
19 MOV D,A
20 INX H
21 JMP LOOP
22 DONE: SHLD 8060H
23 MOV A,E
24 STA 8062H
25 MOV A,D
26 STA 8063H
27 HLT
```

Memory Window:

Address (Hex)	Address	Data
8060	32864	4
8061	32865	0
8062	32866	0
8063	32867	0
8064	32868	0
8065	32869	0
8066	32870	0
8067	32871	0
8068	32872	0
8069	32873	0
806A	32874	0
806B	32875	0
806C	32876	0
806D	32877	0

Assembler Message:

```
0 Program assembled successfully
```

9. Greatest of Two Numbers

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code is as follows:

```
1 LDA 8050H
2 MOV B,A
3
4 LDA 8051H
5 CMP B
6
7 JC FIRST
8
9 STA 8052H
10 HLT
11
12 FIRST: MOV A,B
13 STA 8052H
14 HLT
```

The registers section shows the following values:

Register	Value
A	35
BC	35 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 14
SP	FF FF
Int-Reg	00

The flags section shows the following values:

Flag	Value
S	1
Z	0
AC	0
P	1
C	1

The memory section shows the following values:

Address (Hex)	Address	Data
8050	32848	53
8051	32849	40
8052	32850	53
8053	32851	0
8054	32852	0
8055	32853	0
8056	32854	0
8057	32855	0
8058	32856	0
8059	32857	0
805A	32858	0
805B	32859	0
805C	32860	0
805D	32861	0

The assembler message section shows the following message:

```
0 Program assembled successfully
```

10. Smallest of Two Numbers

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code is as follows:

```
1 LDA 8050H
2 MOV B,A
3
4 LDA 8051H
5 CMP B
6
7 JC SECOND
8
9 MOV A,B
10 STA 8052H
11 HLT
12
13 SECOND: STA 8052H
14 HLT
```

The registers section shows the following values:

Register	Value
A	28
BC	35 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 14
SP	FF FF
Int-Reg	00

The flags section shows the following values:

Flag	Value
S	1
Z	0
AC	0
P	1
C	1

The memory section shows the following values:

Address (Hex)	Address	Data
8050	32848	53
8051	32849	40
8052	32850	40
8053	32851	0
8054	32852	0
8055	32853	0
8056	32854	0
8057	32855	0
8058	32856	0
8059	32857	0
805A	32858	0
805B	32859	0
805C	32860	0
805D	32861	0

The assembler message section shows the following message:

```
0 Program assembled successfully
```

